SBC 201 Chapter 4 Special Detailed Requirements Based on Occupancy and Use — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–18 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 4, source file `Reference\SBC 201 2024\source_reference\Chapter_04 – SPECIAL DETAILED REQUIREMENTS BASED ON OCCUPANCY AND USE.txt`.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **768** independently checkable numeric records (**403** Verified, **365** Verify source). Unresolved OCR and flattened appended tables are listed in the register and are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, fire-engineering report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapter 2 (high-rise definition), Chapter 5 height/area tables, Table 601/602, Chapter 7 ratings, Chapter 9, Chapter 10, Chapter 11, Chapter 16, Chapter 30, SBC 501, SBC 801, ICC A117.1, commentary examples, or a chapter summary. Where Chapter 4 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications. Section **403.1** does not restate the Chapter 2 occupied-floor height; smokeproof-stair **403.5.4** uses **23 m** above the lowest level of fire department vehicle access.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection is not selected. **Section 403.3** requires a system throughout in accordance with **Section 903.3.1.1** (NFPA 13) for high-rise buildings. **NFPA 13R / Section 903.3.1.2** is **not** an equal 403.3 path. Section **420.4** still *points* to **903.2.8** for Group R; that outbound value is not imported.
4. Construction type, Risk Category, Seismic Design Category, building height versus **36 m** and **128 m**, podium garage type (open vs enclosed), atrium, and occupied basement depth versus **9 m** below lowest level of exit discharge are unconfirmed.
5. Mall, Group I, Group H, aircraft, stages, laboratories and other occupancy-only packages in this chapter are omitted from project-use tables. They remain in the internal inventory and coverage counts.
6. Table **406.5.4** and most other appended tables are flattened HTML. Affected cells are **Verify source**. Table **403.2.4** two-cell bond-strength values are readable and are treated as Verified.
7. Fire-partition / horizontal-assembly ratings in Section 708/711, fire-command-center construction in Section 911, smokeproof-enclosure construction in Section 909.20, fire-service-access elevator construction in Section 3007, and domestic-cooking rules in SBC 501 require separate verification.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, height band, construction type, sprinkler branch, podium garage, atrium or basement exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 4 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 4 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 4 application	Required project action
High-rise datum	Occupied floor stated above 23 m; exact height, grade plane and lowest fire-department vehicle access unconfirmed	Selects 403 package; **36 m** fire-service-access elevators; **128 m** FRR-reduction stop, extra risers, dual mains, enclosure impact integrity	
Sprinkler basis	Unconfirmed. High-rise **403.3** requires **903.3.1.1**; **420.4** points to **903.2.8**	13R is not a 403.3 substitute. FRR reductions, fuel-line rating drop and shaft-rating drop all assume the 403.3 system	
Construction type	Unconfirmed; high-rise R-2 typically Type I or II	403.2.1.1 IA→IB and IB→IIA reductions; Type IV-A/IV-B **36 m** dual-main trigger	
Height vs 128 m	Unconfirmed	Extra stair **does not apply** to R-2; riser redundancy, dual street mains, SFRM **48 kN/m²** and enclosure impact still may	
Risk Category	Unconfirmed; conventional R-2 often II	403.2.2 impact integrity applies to Risk Category III/IV at any high-rise height, and to all buildings **more than 128 m**	
Seismic Design Category	Unconfirmed	403.3.3 secondary on-site water (**30 minutes**) applies to SDC C, D, E or F	
Podium / mixed use	Unconfirmed	Shared 403 systems still apply to the tower; garage uses **406**; mall **402** is not assumed	
Parking garage	Unconfirmed: open vs enclosed, private vs public, basement depth	406 geometry; 405 Exception 2 if a sprinklered garage would otherwise be an underground building	

Decision / gap	Current project basis	Why it controls Chapter 4 application	Required project action
Atrium / lobby void	Unconfirmed	404 smoke control, enclosure and glass-protection sprinklers if a connecting void is an atrium	
Occupied basement	Unconfirmed vs **9 m** below lowest LED	405 Type I, two exits, smokeproof stairs; **18 m** two-compartment split	
EVACS / fire command / ERCES	Unconfirmed	403.4.4–403.4.6 send to 907.5.2.2, 911 and SBC 801 §510; values not in this chapter	
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	High-rise smoke removal, smokeproof stairs, FSAE and sprinkler zoning cannot be concluded from Chapter 4 alone	

4. Chapter 4 application

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
401.1	Chapter 4 supplemental scope	Occupancy and construction rules of this code still apply; this chapter adds the occupancy- and use-specific provisions described herein	Any occupancy or use listed in Chapter 4	None stated	Direct	Read 403 and 420 with the rest of the code; do not treat Chapter 4 as a substitute for Chapters 5–10	Verified
402.1 Ex. 1	Mall section not required for R-2 lobby	Foyers and lobbies of Group B, R-1 and R-2 occupancies are **not required** to comply with Section 402	Residential foyer/lobby that is not a covered or open mall building	Buildings that fully comply with other code provisions also need not use 402	Direct	Do not apply mall width, kiosk or plastic-sign rules to the tower lobby unless a true mall building is proposed	Verified

5. High-rise applicability and height bands

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.1	High-rise package trigger	High-rise buildings shall comply with Sections **403.2 through 403.6**	Building classified as high-rise (Chapter 2 definition is outbound)	Airport traffic-control towers 412.2; open parking garages 406.5; Group A-5 portion 303.6; special industrial 503.1.1; H-1; listed H-2/H-3 cases	Direct	Lock occupied-floor height vs lowest fire-department vehicle access on the code data sheet, then apply the 403 package	External verification
403.5.4	Smokeproof-stair height datum	Every required interior exit stairway serving floors **more than 23 m** above the lowest level of fire department vehicle access shall be a smokeproof enclosure	Required interior exit stair serving those floors	Construction of the enclosure is in **909.20** and **1023.12** (not imported)	Direct	Identify which stairs serve floors above the 23 m fire-access datum; do not smokeproof-label podium-only stairs that never serve those floors	Verified
403.6.1	Fire-service elevator height band	Occupied floor **more than 36 m** above the lowest level of fire department vehicle access requires fire-service-access elevators	Tower occupied floor above **36 m** fire-access datum	Capacity and 3007/3002.4 details are in that row	Conditional	Dimension the highest occupied floor vs fire-access grade; if above 36 m, reserve two FSAE cars	Verified
403.2.1.1 / 403.3.1 / 403.5.2	Super-high-rise 128 m band	Several 403 rules change at building height **greater than 128 m**: IA FRR reduction stops; extra sprinkler risers; dual street mains; R-2 extra-stair exemption still holds	Building height vs **128 m**	R-2 extra stair is independently excepted (Section 13)	Conditional	Put **128 m** on the height diagram even if the extra stair is not required for R-2	Verified

6. Construction and fire-resistance reductions

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.2.1	FRR-reduction prerequisite	403.2.1.1 and 403.2.1.2 reductions are allowed only where sprinkler control valves have supervisory initiating devices and water-flow initiating devices **for each floor**	Designer proposes a Table 601 rating reduction	No reduction without the per-floor valve/flow devices	Conditional	Show floor-control assemblies and addressable flow/tamper on the sprinkler schematic before taking a 403.2.1 reduction	Verified
403.2.1.1 Item 1	Type IA element-rating reduction	For buildings **not greater than 128 m** in building height, Type IA building-element ratings may be reduced to Type **IB** ratings	Type IA selected for height/area; height ≤ **128 m**	Columns supporting floors shall **not** be reduced	Conditional	Keep floor-supporting columns on the Type IA column rating; reduce only the other IA elements if this branch is used	Verified
403.2.1.1 Item 2	Type IB element-rating reduction	Type IB building-element ratings may be reduced to Type **IIA** ratings	Type IB selected; occupancy is other than F-1, H-2, H-3, H-5, M and S-1	R-2 is not in the excluded list. Reduction still needs 403.2.1 devices	Conditional	If the tower is Type IB, document the IIA-equivalent element ratings on the construction-type sheet	Verified
403.2.1.1 Item 3	Height/area after reduction	Height and area limits remain those of the building **without** the rating reduction	Any 403.2.1.1 reduction taken	Construction type is not reclassified downward	Conditional	Keep Tables 504/506 on the original type; do not shrink allowable area because elements were reduced	Verified
403.2.1.2	Shaft-enclosure rating reduction	For height **not greater than 128 m**, fire barriers enclosing vertical shafts other than interior exit stairways and elevator hoistways may be reduced to **1 hour** where sprinklers are installed in the shafts at the **top** and at **alternate floor levels**	Secondary shafts (trash, MEP, linen) in a ≤ **128 m** high-rise	Does **not** apply to exit-stair or elevator-hoistway enclosures	Conditional	Detail shaft-head and alternate-floor sprinklers on every reduced-rating shaft; keep exit and lift shafts at their full rating	Verified

7. Stair and elevator enclosure impact integrity

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.2.2	Enclosure integrity trigger	Interior exit-stairway and elevator-hoistway enclosures shall comply with 403.2.2.1–403.2.2.4 for Risk Category **III or IV**, and for **all** buildings **more than 128 m** in building height	Risk Category III/IV high-rise, or any building > **128 m**	Typical Risk Category II R-2 below 128 m is outside this trigger	Conditional	Confirm Risk Category on the structural criteria sheet before specifying ordinary shaft board	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.2.2.1	Soft-body impact	Enclosure panels shall meet or exceed Soft Body Impact Classification **Level 2**, ASTM C1629/C1629M, tested from the **exterior** side of the enclosure	403.2.2 applies	Concrete/masonry deemed to comply (403.2.2.3)	Conditional	Specify Level 2 soft-body shaft board (or concrete/masonry) on the core-wall type	Verified
403.2.2.2	Hard-body impact	Face **not** exposed to the interior of the enclosure: **two** layers of Hard Body **Level 2**, **or one** layer of Hard Body **Level 3**, **or** multiple layers tested in tandem to **Level 3**	403.2.2 applies	Concrete/masonry deemed to comply (403.2.2.3)	Conditional	Note the chosen hard-body build-up on the corridor side of the shaft	Verified
403.2.2.3–403.2.2.4	Masonry or equivalent walls	Concrete or masonry walls satisfy 403.2.2.1 and 403.2.2.2. Other materials may be used if they provide equivalent Soft Body **Level 2** and Hard Body **Level 3** resistance, ASTM C1629/C1629M	Alternative shaft wall proposed	Hard-body Level 3 text is page-split after 403.2.2.4 heading; classification is completed on the next page	Conditional	Prefer concrete/masonry cores, or attach the C1629 report for a board system	Verified

8. Sprayed fire-resistant materials

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.2.3 / Table 403.2.4	SFRM bond strength	SFRM installed throughout the building shall have minimum bond strength **21 kN/m²** for height **up to 128 m**, and **48 kN/m²** for height **greater than 128 m**, measured above the lowest level of fire department vehicle access (Table note a)	High-rise with sprayed fire-resistant materials	Table is concatenated HTML; the two cells are readable. Do not use commentary **7.2 kN/m²**	Direct	Put the governing bond-strength value on the fireproofing specification and mock-up	Verified

9. High-rise sprinklers, risers, water and pumps

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.3	High-rise sprinkler standard	Building shall be equipped throughout with an automatic sprinkler system in accordance with **Section 903.3.1.1** and a secondary water supply where required by 403.3.3	High-rise building under 403	Telecommunications-equipment spaces: detection per 907.2 plus **1-hour** fire barriers or **2-hour** horizontal assemblies, or both. **903.3.1.2 / NFPA 13R is not listed**	Direct	Specify NFPA 13 / 903.3.1.1 throughout the tower; do not use a 13R-only high-rise design	Verified
403.3.1	Extra sprinkler risers	Each sprinkler zone in buildings **more than 128 m** in building height shall be supplied by **not fewer than two** risers. Each riser supplies sprinklers on **alternate** floors. If more than two risers serve a zone, adjacent floors shall not be supplied from the same riser	Building height > **128 m**	Does not apply at or below 128 m	Conditional	Draw two remote risers per zone and alternate-floor take-offs on the riser diagram	Verified
403.3.1.1	Riser location	Sprinkler risers shall be placed in interior exit stairways and ramps remotely located in accordance with **Section 1007.1**	Risers required	1007.1 separation values are not imported	Direct	Route risers in remote exit stairs; coordinate with 403.5.1 enclosure remoteness	External verification
403.3.2	Dual street mains	Required fire pumps shall be supplied from **not fewer than two** water mains in **different streets**, with separate supply piping, where the building is **more than 128 m** high, **or** Type **IV-A / IV-B** and **more than 36 m** high	Those height/construction cases	Two connections to the **same** main are permitted if the main is valved so an interruption can be isolated and supply continues through **not fewer than one** connection	Conditional	Confirm street-main layout with civil; Type IV-A/IV-B towers hit this at 36 m, not 128 m	Verified
403.3.3	Secondary on-site water	Automatic secondary **on-site** water supply, capacity not less than the hydraulically calculated sprinkler demand including hose stream, duration **not less than 30 minutes** per NFPA 13 occupancy-hazard classification, for high-rise buildings in Seismic Design Category **C, D, E or F**	SDC C–F high-rise	Extra fire pump for the secondary supply only if needed for intake pressure. NFPA 13 demand figures are not imported	Conditional	Lock SDC from Chapter 16; size the on-site tank from the NFPA 13 calculation, not from this matrix	Verified
403.3.4	Fire-pump room	Fire pumps shall be located in rooms protected in accordance with **Section 913.2.1**	Fire pump provided	913.2.1 ratings are not imported	Direct	Show the pump room on the life-safety plan and cite 913.2.1 for the enclosure	External verification

10. Detection, alarm, standpipe, communications and fire command

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.4.1	High-rise smoke detection	Smoke detection in accordance with **Section 907.2.13.1**	High-rise building	Detector locations are not in this chapter	Direct	Carry 907.2.13.1 into the fire-alarm scope; do not invent device counts here	External verification
403.4.2	High-rise fire alarm	Fire alarm system in accordance with **Section 907.2.13**	High-rise building	Outbound	Direct	Show a high-rise fire-alarm system on the cause-and-effect matrix citing 907.2.13	External verification
403.4.3	High-rise standpipe	Standpipe system as required by **Section 905.3**	High-rise building	Class and locations are in Chapter 9	Direct	Coordinate standpipe stairs with 403.5 remoteness; do not copy 905 numbers here	External verification
403.4.4	Emergency voice/alarm	Emergency voice/alarm communication system in accordance with **Section 907.5.2.2**	High-rise building	EVACS details are outbound	Direct	Include EVACS in the fire-alarm specification; this chapter does not size speakers	External verification

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.4.5	Emergency responder coverage	In-building two-way emergency responder communication coverage in accordance with **Section 510 of SBC 801**	High-rise building	SBC 801 §510 exceptions are not imported	Direct	Flag ERCES/DAS as a fire-strategy item for SCD; do not design radio coverage from this extract	External verification
403.4.6	Fire command center	Fire command center complying with **Section 911**, in a location approved by the fire code official	High-rise building	911 room size and equipment list are not imported	Direct	Locate a fire command center at an SCD-agreed entry and cite Section 911 for fit-out	External verification

11. Post-fire smoke removal

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.4.7 Item 1	Natural smoke-removal openings	Manually operable windows or panels around the perimeter of **each floor** at **not more than 15 m** intervals; operable area **not less than 4 m² per 15 m** of perimeter	Natural-ventilation smoke-removal option	Exception 1 (**0.2 m²** per sleeping unit) is **Group R-1 only** — do not use for R-2. Exception 2: fixed glazing permitted if fire fighters can clear it	Direct	Schedule panel size and spacing per floor, or use tempered/clearable glass agreed with civil defence	Verified
403.4.7 Item 2	Mechanical smoke-removal rate	Mechanical air-handling providing **one exhaust air change every 15 minutes** for the area involved; return and exhaust moved directly outside with **no recirculation**	Mechanical smoke-removal option	This is post-fire salvage/overhaul ventilation, not Section 909 atrium smoke control	Conditional	If mechanical removal is chosen, show 4 ACH equivalent (one change / 15 min) as a dedicated exhaust mode	Verified
403.4.7 Item 3	Alternative smoke removal	Any other approved design that will produce equivalent results	Designer proposes a non-prescriptive system	Building-official approval required	Conditional	Use only with a documented equivalent and AHJ acceptance	Verified

12. Standby and emergency power

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.4.8	Standby and emergency power	Standby power per **2702** and **3003** for 403.4.8.3 loads. Emergency power per **2702** for 403.4.8.4 loads	High-rise building	Load lists below; durations in Chapter 27 are not imported	Direct	Split standby vs emergency bus on the generator schedule using the two lists	External verification
403.4.8.1	Generator room enclosure	Indoor generator set in a separate room enclosed with **2-hour** fire barriers (Section 707) or horizontal assemblies (Section 711), or both. Manual start and transfer at the fire command center	Generator inside the building	Group I-2 Condition 2 critical-branch start/transfer need not be at the fire command center — not an R-2 exception	Direct	Draw a 2-hour generator room and duplicate start/transfer at the fire command center	Verified
403.4.8.2 Item 1	UL 1489 fuel-line protection	Fuel lines serving an indoor generator, outside the generator room, protected by a UL 1489 system rated **not less than 2 hours**; reduced to **1 hour** where the building is sprinklered throughout per **903.3.1.1**	Fuel-fired indoor generator	High-rise 403.3 already requires 903.3.1.1, so the **1-hour** reduction is available on that branch	Conditional	Specify 1-hour UL 1489 wrap if NFPA 13 throughout is documented	Verified
403.4.8.2 Item 2	Rated assembly around fuel lines	Assembly fire-resistance **not less than 2 hours**, reduced to **1 hour** where sprinklered throughout per **903.3.1.1 or 903.3.1.2**	Fuel-fired indoor generator; rated shaft/chase used instead of UL 1489	Item 2 is the only 403 fuel-line clause that also names 903.3.1.2. Item 3: other approved methods	Conditional	Prefer the 903.3.1.1 / 1-hour chase consistent with 403.3; do not switch the whole tower to 13R to take this drop	Verified
403.4.8.3	Standby power loads	Standby loads: (1) smokeproof-enclosure ventilation and automatic fire detection; (2) elevators; (3) additional standby rules in **1009.4, 3007 or 3008** where elevators serve accessible egress, fire-service access or occupant self-evacuation	High-rise standby generator	3003/3007/3008 kW figures are not imported	Direct	Include smokeproof fans and all elevators on the standby schedule; add FSAE/OEE extra load if those elevators exist	External verification
403.4.8.4	Emergency power loads	Emergency loads: exit signs and egress illumination (Chapter 10); elevator car lighting; EVACS; automatic fire detection; fire alarm; electrically powered fire pumps; fire-command-center power and lighting	High-rise emergency generator	Pickup times are in Chapter 27, not here	Direct	Put these seven loads on the emergency bus, including fire-command lighting	External verification

13. High-rise means of egress

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.5.1	Interior exit-stair remoteness	Separate required interior exit stairways by **not less than 10 m or one-fourth** of the maximum overall diagonal of the building or area served, **whichever is less**. Measure in a straight line between nearest points of the **enclosures**. Where **three or more** interior exit stairways exist, **not fewer than two** shall comply. Interlocking or scissor stairways count as **one**	High-rise required interior exit stairs	Also meet Chapter 10 Section 1007 (values not imported). Do **not** use commentary **9 m**	Direct	Dimension enclosure-to-enclosure clearance on each typical floor; treat scissors as one stair	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.5.2	Additional stair — R-2 exemption	One additional interior exit stairway is required for buildings **other than Group R-2 and their ancillary spaces** that are **more than 128 m** in building height, plus remaining-capacity-with-one-stair-removed. Scissor stairs shall not be that additional stair	Building height > **128 m** in a **non-R-2** occupancy	Exception 1: occupant-evacuation elevators per 3008. Exception 2: portions with highest occupiable floor **less than 128 m**. **R-2 and ancillary amenity spaces are exempt from the extra stair**	Direct	Do not add a third stair solely for 403.5.2 on an R-2 tower; still size stairs to Chapter 10 and 403.5.1	Verified
403.5.3	Stair re-entry unlocking	Stairway doors other than exit-discharge doors may be locked from the stair side if they can be unlocked **simultaneously without unlatching** on a signal from the fire command center	Stair doors locked against re-entry	Unlocking shall not defeat the latch	Conditional	If security locks stairs, show a fire-command unlock circuit that leaves latching intact	Verified
403.5.3.1	Locked-stair communication	Telephone or other two-way communications to an approved constantly attended station at **not less than every fifth floor** in each stairway where doors are locked	403.5.3 locking used	None stated in the numbered code sentence	Conditional	Place a two-way device at least every fifth locked-stair landing	Verified
403.5.4	Smokeproof interior exit stairs	Every required interior exit stairway serving floors **more than 23 m** above the lowest level of fire department vehicle access shall be a smokeproof enclosure in accordance with **909.20** and **1023.12**	Those stairs	Vestibule vs pressurization methods are in 909.20, not imported	Direct	Designate tower exit stairs as smokeproof and send the system design to 909.20	External verification
403.5.5	Luminous egress path markings	Luminous egress path markings shall be provided in accordance with **Section 1025**	High-rise building under 403	This chapter does **not** list occupancy groups. Do not import a commentary A/B/E/I/M/R-1 list	Direct	Apply 1025 to the exit enclosures; verify occupancy scope in Chapter 10, not from this matrix	External verification

14. High-rise elevators

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
403.6	Elevator charging	Elevator installation and operation shall comply with **Chapter 30** and 403.6.1–403.6.2	High-rise building	Chapter 30 car sizes and hoistway protection are not imported	Direct	Send the lift package to Chapter 30; use 403.6.1–403.6.2 for the high-rise extras	External verification
403.6.1	Fire-service-access elevators	Occupied floor **more than 36 m** above the lowest level of fire department vehicle access: **not fewer than two** fire-service-access elevators, **or all elevators, whichever is less**, in accordance with **Section 3007**. Each FSAE capacity **not less than 1600 kg**, and comply with **Section 3002.4**	Occupied floor above the **36 m** fire-access datum	3007 lobby, standpipe-stair adjacency and 3002.4 stretcher dimensions are not imported	Conditional	Provide two 1600 kg FSAE cars (or every car if fewer than two exist) and a 3007 lobby at each FSAE	Verified
403.6.2	Occupant-evacuation elevators	Passenger elevators for general public use **may** be used for occupant self-evacuation where installed in accordance with **Section 3008**	Designer elects OEE	Optional for R-2; not a substitute for R-2 extra-stair (that stair is already exempt)	Conditional	If OEE is offered, apply 3008 to **all** public passenger lifts; do not import 3008 geometry here	External verification

15. Group R unit separations and domestic cooking

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
420.1	Group R charging	Groups I-1, R-1, R-2, R-3 and R-4 shall comply with 420.1 through 420.5 and other applicable provisions	Group R-2 building	420.6–420.9 are I-1 only and are omitted here	Direct	Apply unit separations, 420.4/420.5 pointers and 420.10 cooking on every R-2 floor	Verified
420.2	Dwelling/sleeping-unit walls	Walls separating dwelling units, separating sleeping units, and separating dwelling or sleeping units from other contiguous occupancies shall be **fire partitions** in accordance with **Section 708**	R-2 units in the same building	708 ratings (including any sprinkler reduction) are not imported	Direct	Draw unit-demising and unit-to-corridor/other-occupancy walls as 708 fire partitions; read the hour rating from Chapter 7	External verification
420.3	Dwelling/sleeping-unit floors	Floor assemblies separating dwelling units, separating sleeping units, and separating dwelling or sleeping units from other contiguous occupancies shall be **horizontal assemblies** in accordance with **Section 711**	R-2 stacked units	711 ratings are not imported	Direct	Rate unit-to-unit and unit-to-podium floors as 711 horizontal assemblies	External verification
420.4	Group R sprinkler pointer	Group R occupancies shall be equipped throughout in accordance with **Section 903.2.8**. Quick-response or residential sprinklers per **903.3.2**	Group R	High-rise **403.3** already requires **903.3.1.1**. Do not import a 13R option from 903.2.8 into the tower	Direct	Keep the tower on NFPA 13 / 903.3.1.1; use 903.3.2 for residential/QR heads where Chapter 9 requires them	External verification
420.5	R-2 alarm and smoke-alarm pointer	Fire alarm and smoke alarms for R-2 in accordance with **Section 907.2.9**. Single- or multiple-station smoke alarms for R-2 in accordance with **Section 907.2.11**	Group R-2	Device locations and interconnection are in Chapter 9	Direct	Include 907.2.9 system and 907.2.11 unit smoke alarms in the fire-alarm specification	External verification
420.10	Group R domestic cooking	Cooking appliances used for domestic cooking operations in Group R shall be in accordance with **Section 917.2 of SBC 501**	Dwelling-unit kitchens and similar domestic cooking	Hotel restaurants / commercial cooking are outside this pointer. Dormitory 420.11 is omitted unless a college-dorm program is opened	Direct	Specify dwelling kitchens to SBC 501 §917.2; do not use I-1 420.8–420.9 rules on apartments	External verification

16. Podium and parking garage

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
406.2.2	Garage clear height	Clear height of each floor level in vehicle and pedestrian traffic areas **not less than 2100 mm**	Parking garage present	Mechanical-access open-parking tier may be lower where approved. Accessible van height is in Chapter 11, not here. Fuel-canopy height is 406.7.2	Conditional	Set typical garage storey height to preserve **2100 mm** clear under beams, signs and sprinklers	Verified
406.2.4	Garage floor surface	Concrete or similar approved noncombustible, nonabsorbent floor; parking area sloped to a drain or the main vehicle entry	Parking garage	Exception 1: asphalt at **ground level** for public garages and private carports. Exception 2 (repair-garage CRF **0.45 W/cm²**) is not a typical R-2 podium finish	Conditional	Slope basement/upper garage slabs to drains; limit asphalt to grade-level if used	Verified
406.2.5	Garage/sleeping openings	Openings between a motor-vehicle-related occupancy and a room used for **sleeping purposes** shall **not** be permitted	Any garage adjoining sleeping rooms	Applies to public and private garages	Conditional	Eliminate windows/doors from garage into bedrooms; use a non-sleeping lobby if a unit adjoins the garage	Verified
406.2.7	EV charging listing	EV charging equipment listed to **UL 2202**; EV supply equipment listed to **UL 2594**; electrical per NFPA 70 / SBC 401; accessibility per **Section 1107**	EV charging provided in the garage	Accessible EVCS geometry is in Chapter 11 / A117.1, not imported	Conditional	Put UL 2202/2594 on the EVCS specification and send accessible stalls to Chapter 11	External verification
406.2.9.1	Ignition-source elevation	Ignition sources in garages elevated **not less than 450 mm** above the floor on which the appliance rests	Fuel-fired or sparking equipment in the garage	Not required for appliances listed flammable-vapor-ignition-resistant. Rooms communicating with a private garage are treated as part of that garage	Conditional	Mount water heaters, dryers and similar equipment ≥ **450 mm** or use FVIR-listed units	Verified
406.2.9.1.1	Fuel-fired appliance vestibule	Connection of a parking garage to a room with a fuel-fired appliance shall be by a **two-doorway** vestibule, unless ignition sources are elevated per 406.2.9	Public/parking garage next to a plant room	Single door permitted with the elevation option. 406.2.9.2 / 406.2.9.3 installations are excepted	Conditional	Provide a vestibule to boiler/generator rooms opening to the garage, or elevate the ignition source	Verified
406.2.9.2	Public-garage appliance height	Appliances in public garages **not less than 2400 mm** above the floor; if vehicles can pass under, **not less than 300 mm** higher than the tallest vehicle door opening	Public parking garage	Exception: impact-protected appliances installed per 406.2.9.1 and NFPA 30A	Conditional	Hang garage heaters/AHUs at **2400 mm** or behind vehicle-impact protection	Verified
406.3.1	Private-garage area cap	Each private garage **not greater than 100 m²**; multiple private garages separated by **1-hour** fire barriers (707) or **1-hour** horizontal assemblies (711), or both	Private (Group U) garages rather than public S-2	A typical podium public garage is 406.4, not this cap	Conditional	Use 406.4/406.5/406.6 for a shared podium garage; apply 100 m² only to true private garages	Verified
406.3.2.1	Private garage / dwelling separation	**12.5 mm** gypsum on the garage side; garage beneath habitable rooms: **16 mm** Type X (or equivalent) plus **12.5 mm** on supporting structure. Doors: solid wood or honeycomb steel **not less than 35 mm**, or **20-minute** 716.2.2.1 doors; self-closing and self-latching	Private garage adjacent to a dwelling unit	Not a substitute for 420.2/420.3 in the tower. 406.2.5 still bans openings into sleeping rooms	Conditional	Use this build-up only for townhouse-style private garages, not for the public podium	Verified
406.4.2	Vehicle barriers	Vehicle barriers **not less than 850 mm** high where the drop from drive lane or parking space to the surface below is **greater than 300 mm**; loading per **Section 1607.10**	Public parking garage with edge drop > **300 mm**	Not required in mechanical-access vehicle storage compartments. 1607.10 kN value is not imported	Conditional	Show 850 mm vehicle barriers at every garage edge and ramp drop; send loads to the structural engineer	Verified
406.4.3	Parking-ramp slope	Vehicle ramps used for vertical circulation **and** parking shall **not exceed 1:15 (6.67 percent)**. Vehicle ramps are not required exits unless pedestrian facilities are provided	Parking on the ramp	Steeper ramps cannot be the pedestrian exit	Conditional	Keep parked ramps at **1:15** or flatter; provide separate pedestrian exits	Verified
406.5.2	Open-garage ventilation openings	Openings on **two or more** sides; opening area **not less than 20 percent** of each tier's perimeter wall area; aggregate opening length **not less than 40 percent** of the tier perimeter; interior walls **not less than 20 percent** open	Open parking garage classification	Exception: 40 percent perimeter distribution not required where openings are uniformly distributed on **two opposing** sides (20 percent area still applies)	Conditional	If claiming "open" garage, calculate opening area and length per tier before dropping mechanical ventilation	Verified
406.5.2.1	Below-grade open-garage well	Where below-grade openings provide the required natural ventilation, outside horizontal clear space shall be **one and one-half** times the depth of the opening, maintained from grade to the lowest required opening	Open garage with openings below surrounding grade	Depth-to-well ratio is the check	Conditional	Size light-well width ≥ **1.5 ×** opening depth	Verified
406.5.4.1	Single-use open garage / spiral tier	Exclusive parking use may use **Table 406.5.4**. Grade-level office/waiting/toilets **not more than 100 m²** need not be separated. Continuous spiral floor: each **2900 mm** of height (or portion) is a tier	Stand-alone open garage, not mixed with the tower	Table 406.5.4 cells are flattened — **Verify source**. Mixed podium garages stay on Chapter 5 via 406.5.4	Conditional	Do not use Table 406.5.4 for a mixed R-2 podium; verify the published table only for a single-use open garage	Verify source
406.5.5	Open-garage area/height increases	Sides open on **three-fourths** of the perimeter: **25 percent** area increase and **one** extra tier. Entire perimeter open: **50 percent** area and **one** extra tier. "Open" side: openings **not less than 50 percent** of interior side area, height used in the calculation **not more than 2.15 m**	Single-use open garage seeking Table 406.5.4 increases	Type II all-sides-open unlimited-area branch (height **not exceed 23 m**; portions within **6 m** of openings; courts **60 m**) is page-split — treat those two distance tokens as **Verify source**	Conditional	Apply 25/50 percent increases only after opening-area diagrams; verify the Type II 6 m / 60 m page-split before using unlimited area	Verify source

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
406.5.7	Attendant-only garage stairs	Where persons **other than parking attendants** are not permitted: **not fewer than two** exit stairways, each **not less than 900 mm** wide	Attendant-only / mechanical-access open garage	Public-use garages use Chapter 10 (not imported)	Conditional	For a public podium garage, size stairs from Chapter 10, not the 900 mm attendant branch	Verified
406.6.2	Enclosed-garage ventilation	Mechanical ventilation and exhaust in accordance with **Chapters 4 and 5 of SBC 501**	Enclosed public parking garage	Exception: accessory to one- and two-family dwellings	Conditional	Send enclosed-podium ventilation rates to SBC 501; do not invent ACH here	External verification
406.6.3	Enclosed-garage sprinklers	Enclosed parking garage sprinklered in accordance with **Section 903.2.10**	Enclosed public garage	903.2.10 fire-area thresholds are not imported. High-rise 403.3 already requires 13 throughout the building	Conditional	Keep enclosed podium parking on the building NFPA 13 system; confirm 903.2.10 still met	External verification
406.6.4.1–406.6.4.3	Mechanical-access enclosed garage	**2-hour** separation from other occupancies; mechanical smoke removal per **910.4**; fire-control room **not less than 4.65 m²** with exterior fire-service access	Mechanical-access enclosed (rack) garage	Typical ramp podium is 406.6, not 406.6.4	Conditional	If a puzzle/rack garage is added, apply 2-hour box, 910.4 exhaust and a 4.65 m² fire-control room	Verified

17. Atriums

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
404.1	Atrium charging	Sections 404.1–404.11 apply to buildings containing atriums. Atriums are not permitted in Group H	Connecting vertical void treated as an atrium	Vertical openings that comply with 712.1.1–712.1.3 and 712.1.9–712.1.14 need not use 404	Conditional	Decide atrium vs 712 opening on the lobby section; do not skip 404 if 712.1.7 (atrium) is the chosen path	Verified
404.3	Atrium building sprinklers	Approved automatic sprinkler system throughout the **entire building**	Atrium present	Exception 1: adjacent/above areas separated by **2-hour** barriers/horizontal assemblies. Exception 2: atrium ceiling **more than 16 m** above the floor — ceiling sprinklers not required (do not use commentary 16.5 m)	Conditional	Sprinkler the whole building if an atrium is used; omit only the high ceiling or 2-hour-separated wings	Verified
404.5	Atrium smoke control	Smoke control in accordance with **Section 909**	Atrium connecting stories	Exception 1: smoke control not required for a **two-story** atrium other than I-2 / I-1 Condition 2. Exception 2: more than two stories if only the **two lowest** stories are open and floors above are shaft-separated per 713.4	Conditional	Provide 909 smoke control unless the two-story or shafted-upper-floor exception is demonstrated	External verification
404.6	Atrium enclosure	Separate atrium from adjacent spaces by a **1-hour** fire barrier (707) or horizontal assembly (711), or both	Atrium present	Glass-wall exception: sprinklers **100–300 mm** from glass, spacing **not greater than 1800 mm**, both sides (or room side only if no atrium walkway). Glass-block **3/4-hour**. Up to **three** floors may be open if included in the smoke-control design. No enclosure where 404.5 smoke control is not required	Conditional	Default to 1-hour atrium walls, or detail the 100–300 mm glass sprinkler grid	Verified
404.8	Atrium interior finish	Walls and ceilings of the atrium **not less than Class B**; sprinklers shall **not** reduce the class	Atrium surfaces	Table 803.13 reductions are not allowed here	Conditional	Specify Class B (or better) atrium wall/ceiling finish with no sprinkler trade-down	Verified
404.10	Exit stair in the atrium	Entry is the closest riser; access from **minimum of two** directions; remoteness to an enclosed 1023.2 stair per 1007.1.1; travel measured to the closest riser; **not more than 50 percent** of exit stairways in the same atrium	Interior exit stair placed in the atrium	1007.1.1 distances not imported	Conditional	If a feature stair is an exit, keep at least half the exits in enclosed stairs and show two-way approach to the first riser	Verified
404.4 / 404.7 / 404.9 / 404.11	Atrium outbound pointers	Fire alarm **907.2.14**; smoke-control standby power **909.11**; travel distance **1017**; discharge through atrium **1028**	Atrium present	Values in those sections are not imported	Conditional	List 907.2.14, 909.11, 1017 and 1028 as atrium coordination items	External verification

18. Underground buildings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
405.1	Underground-building trigger	405.2–405.9 apply where a floor used for **human occupancy** is **more than 9 m** below the finished floor of the lowest level of exit discharge	Occupied basement / below-grade amenity	Exception 2: **parking garages** with sprinklers per 405.3. Also 1–2 family 903.3.1.3; transit; assembly seating; lowest storey ≤ **140 m²** and occupant load **less than 10**; limited-use plant rooms	Conditional	Measure occupied-floor depth vs LED. Sprinklered parking alone does not make the building a 405 underground building	Verified
405.2	Underground construction type	Underground portion shall be **Type I** construction	405 applies	Above-grade portion follows Chapters 5/6	Conditional	Set below-grade occupied structure to Type I	Verified
405.3	Underground sprinklers	Highest level of exit discharge serving the underground portions, and all levels below, sprinklered per **903.3.1.1**; water-flow and control valves supervised per **903.4**	405 applies	Consistent with high-rise 403.3 NFPA 13	Conditional	Extend NFPA 13 from LED down through all below-grade occupied levels	External verification
405.4.1	Deep-level two-compartment split	Floor **more than 18 m** below lowest LED: divide into **not fewer than two** compartments of approximately equal size, from LED down	Occupied floor > **18 m** below LED	Lowest storey need not be compartmented if area **not greater than 140 m²** and occupant load **less than 10**	Conditional	If a deep occupied basement exists, split it with a smoke barrier into two roughly equal compartments	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
405.4.2–405.4.3	Compartment independence	Smoke barrier per **709**; penetrations limited to plumbing/electrical with **714** firestopping; doors 716, smoke-activated, NFPA 105. Independent air supply/exhaust per compartment. Each compartment has direct elevator access, or a smoke-barrier lobby if the elevator serves more than one compartment	405.4.1 compartmentation required	709/714/716 values not imported	Conditional	Give each deep compartment its own HVAC and either its own lift or a 709 elevator lobby	External verification
405.5 / 405.6	Underground smoke control and alarm	Smoke control per **909**; where compartmented, each compartment has an independent automatically activated, manually operable system. Fire alarm where required by **907.2.18** and **907.2.19**	405 applies	909 design criteria not imported	Conditional	Add below-grade 909 smoke control and the 907.2.18/19 alarm scope	External verification
405.7.1	Underground exit count	Each floor: **not fewer than two** exits. Where 405.4 compartmentation is required: **not fewer than one** exit **and not fewer than one** exit-access doorway into the adjoining compartment	405 applies	Chapter 10 may require more	Conditional	Put an exit in each deep compartment plus a cross-door to the other compartment	Verified
405.7.2	Underground smokeproof stairs	Every required stair serving floors **more than 9 m** below the finished floor of its LED shall be a smokeproof enclosure per **1023.12**	Those stairs	909.20 construction not imported	Conditional	Smokeproof any stair that climbs more than 9 m from a below-grade occupied floor to its LED	External verification
405.8 / 405.9	Underground power and standpipe	Standby: smoke control; smokeproof ventilation/detection; elevators per 3003. Emergency: EVACS, fire alarm, detection, car lighting, egress/exit-sign lighting, fire pumps. Standpipe throughout per **905**	405 applies	May overlap 403.4.8 loads in a high-rise with a deep basement	Conditional	Combine 403 and 405 load lists on the generator schedule; extend standpipe to the lowest occupied level	External verification

19. Project-use controls

1. Use **Verified** rows for drawing and specification checks after the row trigger is confirmed (high-rise, garage type, atrium, occupied depth).
2. Treat **Verify source** rows (flattened Table 406.5.4, 406.5.5 page-split distances) as hold points. They are not design-release values.
3. Do not import 708/711 hours, 907 device layouts, 909.20 vestibule sizes, 911 fire-command dimensions, 1025 marking details, 3007 lobby size, or SBC 501 cooking clearances into issued drawings from this matrix.
4. Keep **NFPA 13 / 903.3.1.1** as the high-rise sprinkler path (403.3). Do not treat 13R as satisfying 403.3 even though 420.4 points at 903.2.8.
5. Do not apply 403.5.2’s extra stair to Group R-2 or its ancillary amenity spaces.
6. Do not give R-2 the 403.4.7 Exception 1 **0.2 m²** hotel venting allowance.
7. Do not use commentary **9 m** in place of code **10 m** for 403.5.1, or commentary **7.2 kN/m²** in place of Table 403.2.4.
8. Omit mall (402), I-2/I-3, H, aircraft, stages, laboratories, dormitory cooking (420.11) and I-1 smoke-compartment cooking unless the gap register is later opened for those uses.
9. Record high-rise height bands, sprinkler standard, garage type and atrium/basement decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

20. Coverage summary

Internal inventory of the attached Chapter 4 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 768
- **Verified:** 403
- **Verify source:** 365

Counts by top-level section

Top-level section	Records
401	0
402	42
403	43
404	12
405	11
406	73
407	29
408	23
409	6
410	26
411	4
412	141

Top-level section	Records
413	2
414	116
415	131
416	1
417	3
418	3
419	3
420	10
421	0
422	7
423	6
424	15
425	0
426	13
427	15
428	33

Appended-table coverage

Appended table	Records	Verify source records
Table 403.2.4	2	0
Table 406.5.4	20	20
Table 412.2.1.1	5	5
Table 412.3.6	81	81
Table 412.6	24	24
Table 414.2.2	36	36
Table 414.2.5(1)	40	40
Table 414.2.5(2)	20	20
Table 414.5.1	15	15
Table 415.6.5	26	26
Table 415.11.1.1.1 (incl. continuation)	66	66
Table 428.3	30	30

No CS.md exists for this chapter. Commentary figures (including Figure 403.5.1's **9 m** example) were not inventoried and were not used as values.

21. Unresolved-source register

Hold points for the 365 **Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
402.8.2.2 OLF range	Code text reads occupant-load factor **not less than 30** and **shall not exceed 50**; SI mall practice and the following commentary use **2.8 / 4.65**. Not adopted	Mall 402 is omitted from project-use. If a mall is later added, verify the published SI OLF bounds before using Equation 4-1
Table 406.5.4	Flattened HTML; 20 area/tier cells not safely segmented	Mixed R-2 podiums use Chapter 5 via 406.5.4, not this table. Verify published cells only for a single-use open garage
406.5.5 Type II unlimited-area distances	Page-split after **2.15 m**; published continuation gives **6 m** horizontal and **60 m** courts, which conflict with the commentary figures on the same pages	Do not release Type II unlimited-area open-garage geometry until the published distances are checked
Table 412.2.1.1	Concatenated control-tower heights	Aircraft occupancy omitted from project-use
Table 412.3.6	Flattened hangar fire-area / construction grid plus **8.5 m** door-height note	Aircraft occupancy omitted
Table 412.6	Flattened aircraft-manufacturing travel-distance grid	Aircraft occupancy omitted
Table 414.2.2	Flattened control-area floor / percentage / rating grid	Hazardous-material control areas omitted unless an H use is added
Table 414.2.5(1)	Flattened MAQ indoor/outdoor cells and sprinkler-cabinet increase notes	Do not reconstruct kg/L MAQs from concatenated HTML
Table 414.2.5(2)	Concatenated flammable-liquid MAQ columns and density footnotes	Verify published litres and Ordinary Hazard densities before any retail MAQ
Table 414.5.1	Flattened explosion-control Required/Not Required grid	H occupancy omitted
Table 415.6.5	Flattened detached-building quantity triggers	H occupancy omitted
Table 415.11.1.1.1 (both pages)	Flattened H-5 fabrication density cells; footnote **255 m³** NTP	H-5 omitted
Table 428.3	Flattened laboratory-suite percentage / count / rating grid	Higher-education laboratories omitted